- Germany until March 2011 obtained one-quarter of its electricity from nuclear energy, using 17 reactors.
- A coalition government formed after the 1998 federal elections had the phasing out of nuclear energy as a feature of its policy. With a new government in 2009, the phase-out was cancelled, but then reintroduced in 2011. Eight reactors were shut down immediately, and all were scheduled to close by the end of 2022.
- In October 2022, the Chancellor decided that Germany's three remaining nuclear power reactors

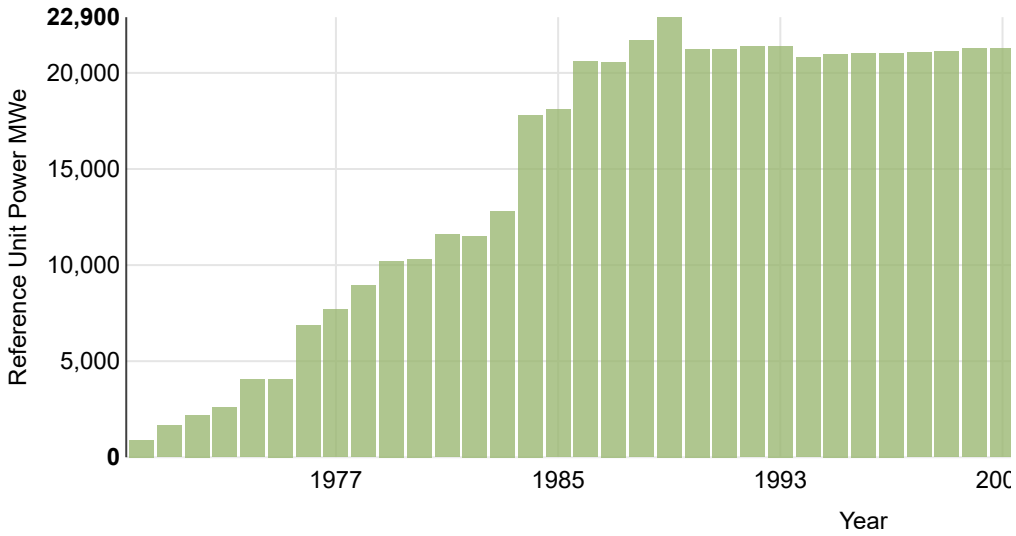
Privacy - Terms

would keep operating until mid-April 2023 to offset reduced gas supply from Russia.

- In May 2025 under Chancellor Friedrich Merz, Germany abandoned its opposition to recognizing nuclear power as equivalent to renewables in EU climate policy.
- Germany has some of the lowest wholesale electricity prices in Europe and some of the highest retail prices, due to its energy policies. Taxes and surcharges account for more than half the domestic electricity price.

0	0	36
OPERABLE REACTORS	REACTORS UNDER CONSTRUCTION	REACTORS SHUTDOWN
0 MWe	0 MWe	26,411 MWe

Operable nuclear power capacity



Electricity sector

Total generation (in 2024): 510 TWh
Generation mix: wind 139 TWh (27%); coal 120 TWh (24%); natural gas 89.1 TWh (17%); solar 74.3 TWh (15%); biofuels &

waste 52.4 TWh (10%); hydro 28.9 TWh (6%); oil 4.9 TWh (1%)

Import/export balance: 26.3 TWh net export (81.7 TWh imports, 55.4 TWh exports)

Total consumption: 454 TWh in 2023

Per capita consumption: c. 5400 kWh in 2023

Source: International Energy Agency and The World Bank.

Exports were mainly to Austria, Netherlands, Poland and Czech Republic, with net imports from France. Germany is one of the biggest importers of gas, coal and oil worldwide, and has few domestic resources apart from lignite and renewables (see later section). The preponderance of coal makes the country Europe's biggest emitter of carbon dioxide.

Generating capacity at the end of 2020 was 234 GWe, comprising 62.2 GWe wind, 53.7 GWe solar, 51.3 GWe coal, 32.8 GWe natural gas, 10.8 GWe hydro, 8.1 GWe nuclear, 3.6 GWe oil, 11.2 GWe other (IEA figures[a]). Total capacity has increased by nearly 140% since 1990 to give only 11% more power; less than 30% of production comes from wind and solar, from about half of total capacity.

German energy association BDEW (*Bundesverband der Energie- und Wasserwirtschaft*, German Association of Energy and Water Industries) has forecast electricity demand in 2030 of 700 TWh to cater for 14 million electric vehicles, 15 GW of electrolyzer capacity using 30 TWh and 6 million heat pumps. Official projections are for 645-665 TWh in 2030.

Nuclear power industry

Shutdown reactors in Germany

Reactor Name	Model	Reactor Type	Reference Unit Power (MWe)	Permanent Shutdown
AVR Juelich	Pebble bed reactor prototype	HTGR	13	1988-12-31
Biblis A	PWR	PWR	1167	2011-08-06
Biblis B	PWR	PWR	1240	2011-08-06

Reactor Name	Model	Reactor Type	Reference Unit Power (MWe)	Permanent Shutdown
Brokdorf	PWR	PWR	1410	2021-12-31
Brunsbuettel	BWR-69	BWR	771	2011-08-06
Emsland	Konvoi	PWR	1335	2023-04-15
Grafenrheinfeld	PWR	PWR	1275	2015-06-27
Greifswald 1	VVER V-230	PWR	408	1990-12-18
Greifswald 2	VVER V-230	PWR	408	1990-02-14
Greifswald 3	VVER V-230	PWR	408	1990-02-28
Greifswald 4	VVER V-230	PWR	408	1990-06-02
Greifswald 5	VVER V-213	PWR	408	1989-11-24
Grohnde	PWR	PWR	1360	2021-12-31
Gundremmingen A	BWR-1	BWR	237	1977-01-13
Gundremmingen B	BWR-72	BWR	1284	2017-12-31
Gundremmingen C	BWR-72	BWR	1288	2021-12-31
HDR Grosswelzheim	Superheated steam reactor	BWR	25	1971-07-20
Isar 1	BWR-69	BWR	878	2011-08-06
Isar 2	Konvoi	PWR	1410	2023-04-15
KNK II	Prototype	FBR	17	1991-08-23
Kruemmel	BWR-69	BWR	1346	2011-08-06
Lingen	BWR with fossil fuel-fired superheater	BWR	183	1977-01-05
Muelheim Kaerlich	PWR	PWR	1219	1988-09-09
Mzfr	-	PHWR	52	1984-05-03
Neckarwestheim 1	PWR	PWR	785	2011-08-06
Neckarwestheim 2	Konvoi	PWR	1310	2023-04-15
Niederaichbach	pressure tube reactor	HWGCR	100	1974-07-31

Reactor Name	Model	Reactor Type	Reference Unit Power (MWe)	Permanent Shutdown
Obrigheim	PWR	PWR	340	2005-05-11
Philippsburg 1	BWR-69	BWR	890	2011-08-06
Philippsburg 2	PWR	PWR	1402	2019-12-31
Rheinsberg	VVER-70	PWR	62	1990-06-01
Stade	PWR	PWR	640	2003-11-14
THTR-300	Pebble bed reactor	HTGR	296	1988-09-29
Unterweser	PWR	PWR	1345	2011-08-06
VAK Kahl	BWR	BWR	15	1985-11-25
Wuergassen	BWR	BWR	640	1994-08-26

Nuclear industry development

By the start of 2011, the country's 17 nuclear power reactors, comprising 15% of installed capacity, supplied more than one-quarter of the electricity (133 TWh net in 2010). Many of the units were large (the 17 totalled over 20 GWe), and the last came into commercial operation in 1989. All units were built by Siemens-KWU. A further PWR had not operated since 1988 because of a licensing dispute. This picture changed following the March 2011 Fukushima Daiichi accident, with the operating fleet being reduced to nine reactors with 12 GWe capacity by the end of 2011, and to just three reactors with 4 GWe capacity by January 2022 (see later sections). The country's final three reactors shut down in April 2023.

Responsibility for licensing the construction and operation of all nuclear facilities is shared between the federal and Länder governments, which confers something close to a power of veto to both.

When Germany was reunited in 1990, all the Soviet-designed reactors in the east were shut down for safety reasons and are being decommissioned. These comprised four operating VVER-440s, a fifth one under construction and a small older VVER reactor.

In 2000 the European Commission approved the merger of two of Germany's biggest utilities, Veba and Viag, to form E.ON, which owned or had a stake in 12 of the country's 19 nuclear reactors which were operating then. In 2016 E.ON spun off Uniper, which

was to take over all its nuclear assets in 2016, but in the event left German nuclear plants with E.ON.

The Federal Ministry of Economics & Technology (BMWi) implements national energy policy.

Nuclear power policy

German support for nuclear energy was very strong in the 1970s following the oil price shock of 1974, and as in France, there was a perception of vulnerability regarding energy supplies. However, this policy faltered after the Chernobyl accident in 1986, and the last new nuclear power plant was commissioned in 1989. Whereas the Social Democratic Party (SPD) had affirmed nuclear power in 1979, in August 1986 it passed a resolution to abandon nuclear power within ten years.

The Grohnde nuclear power plant opened in 1989 and was shut down in 2021

The most immediate effect of this change of policy was to terminate R&D on both the high-temperature gas-cooled and fast breeder reactors after some 30 years of promising work, since much of the work was in North Rhine-Westphalia, which was governed by the SPD. A Christian Democrat (CDU) federal government then maintained support for existing nuclear power generation nationally until defeated in 1998.

In October 1998 a coalition government was formed between the Social Democratic Party (SPD) and the Green Party, the latter having polled only 6.7% of the vote. As a result, these two parties agreed to change the law to phase out of nuclear power. Long drawn-out "consensus talks" with the electric utilities were intended to establish a timetable for phase out, with the Greens threatening unilateral curtailment of licences without compensation if agreement was not reached. All operating nuclear plants then had unlimited licences with strong legal guarantees.

In June 2000 a compromise was announced which saved face for the government and secured the uninterrupted operation of the nuclear plants for many years ahead. The agreement, while limiting plant lifetime to some degree, averted the risk of any federally-enforced plant closures during the term of that government.

In particular, the agreement put a cap of 2623 billion kWh on lifetime production by all 19 operating reactors, equivalent to an average lifetime of 32 years (less than the 35 years sought by industry). Two key elements were a government commitment to respect the rights of utilities to operate existing plants, and a guarantee that this operation and related waste disposal will be protected from any "politically-motivated interference".

Other elements included: a government commitment not to introduce any "one-sided" economic or taxation measures, a recognition by the government of the high safety standards of German nuclear plants and a guarantee not to erode those standards, the resumption of spent fuel transports for reprocessing in France and UK for five years or until contracts expire, and maintenance of two waste repository projects (at Konrad and Gorleben).

In June 2001 the leaders of the 'Red-Green' coalition government and the four main energy companies signed an agreement to give effect to this 2000 compromise. The companies' undertaking to limit the operational lives of the reactors to an average of 32 years meant that two of the least economic ones – Stade and Obrigheim – were shut down in 2003 and 2005 respectively, and the one non-operational reactor (Mülheim-Kärlich, 1219 MWe) commenced decommissioning in 2003. Brunsbüttel was shut down in 2007, as was Krümmel, apart from a few weeks operation in 2009. The agreement also prohibited the construction of new nuclear power plants for the time being and introduced the principle of on-site storage for used fuel.

The agreement was a pragmatic compromise which limited political interference while providing a basis and plenty of time for formulation of a national energy policy. An industry leader reminded his government that "Reliable and cost-effective energy supply must remain an important component of German economic policy". Some speculation centred on the future of the agreement and the revised Atomic Energy Act which followed it under any new government. Parliamentary opposition party leaders said that they would reverse the decision when they could – in the event, eight years later*.

* A major element in the federal government's war of attrition through 1999-2000 against the nuclear utilities was a law retrospectively to tax funds amounting to DM 50 billion which have been contributed by electricity users and set aside in trust as provision for waste management, decommissioning nuclear power plants and rehabilitating lignite mines. Early in 1999 industry promptly served notice that this would be vigorously contested as "a blatant breach of German constitutional rights and legal principles", as depletion of these funds by some DM 25 billion through the tax will leave future generations liable for much of the future costs. However, despite a scathing attack on it by the Chancellor, Mr Schroeder, in cabinet, the tax measure was approved by the upper house on the last day that the new coalition enjoyed a majority there.

Utilities wanted to extend the lifetimes of all reactors initially to 40 years (from average 32 years) and then individually seeking extensions to 60 years as in the USA.

The new Christian Democrat (CDU) and Liberal Democrat (FDP) coalition government elected in September 2009 was committed to rescinding the phase-out policy, but the financial terms took a year to negotiate. If reactor lifetimes were extended from average 32 years to 60 years, the four operating companies would have reaped additional gross profit of €100 billion or more, and the government was keen to secure more than half of this – much more than its extra tax revenue.

In September 2010 a new agreement was reached, to give eight-year licence extensions (from the dates agreed in 2001) for reactors built before 1980, and 14-year extensions for later ones. The price exacted for this was several new measures: a fuel tax of €145 per gram of fissile uranium or plutonium fuel for six years, yielding €2.3 billion per year (about 1.6 c/kWh), payment of €300 million per year in 2011 and 2012, and €200 million 2013-16, to subsidize renewables and for funding rehabilitation at the Asse salt mine

waste repository. A tax of 0.9 c/kWh for the same purpose would follow after 2016. However, utilities could reduce their contribution to renewables if safety upgrades to particular individual nuclear plants cost more than €500 million. At the end of October these measures were confirmed by parliamentary vote on two amendments to Germany's Atomic Energy Act, and this was confirmed in the upper house in November 2010.

All these arrangements were thrown into doubt when in March 2011 the government declared a three-month moratorium on nuclear power plans, in which checks would take place and nuclear policy would be reconsidered. Chancellor Angela Merkel decreed that the country's nuclear power reactors which began operation in 1980 or earlier should be immediately shut down. Those units then closed and were joined by another unit already in long-term shutdown, making a total of 8336 MWe offline under government direction, about 6.4% of the country's generating capacity. This decision was not based on any safety assessment, and did not result in removal of the nuclear fuel tax.

The reactors affected were Biblis A, Neckarwestheim 1, Brunsbuettel, Biblis B, Isar 1, Unterweser, Phillippsburg 1. Already in a long-term shutdown was Kruemmel and this was included despite having started up in 1984.

In May 2011 the Reaktor-Sicherheitskommission (RSK, Reactor Safety Commission) reported that all German reactors were basically sound, and safe. It had reviewed all 17 reactors and evaluated their robustness with respect to natural events affecting the plants, station blackouts and failure of the cooling system, precautionary and emergency measures as well as man-made events affecting the plant, e.g. plane crashes.

However, despite this safety assurance, on 30 May 2011, after increasing pressure from anti-nuclear federal states, the government decided to revive the previous government's phase-out plan and close all reactors by 2022 but without abolishing the fuel tax, thus reneging on the new fuel tax trade-off. The Bundestag passed the measures by 513 to 79 votes at the end of June, and the Bundesrat vote on 8 July confirmed this. Both houses of parliament approved construction of new coal and gas-fired plants despite claiming to retain its carbon dioxide emissions reduction targets, as well as expanding wind energy. This policy of replacing nuclear power with extra fossil fuel capacity and vastly expanding highly-subsidized renewables is known as the *Energiewende* (see Box1).

Box 1: *Energiewende*

The *Energiewende* policy, detailed in the companion Energiewende information page, has proved controversial and expensive. An introductory paragraph in a July 2014 report by *Finadvice* for the Edison Electric Institute and European clients (see later section with details of this) stated[1]:

Over the last decade, well-intentioned policymakers in Germany and other European countries created renewable energy policies with generous subsidies that have slowly revealed themselves to be unsustainable, resulting in profound, unintended consequences for all industry stakeholders. While these policies have created an impressive roll-out of renewable energy resources, they have also clearly generated disequilibrium in the power markets, resulting in significant increases in energy prices to most users, as well as value destruction for all stakeholders: consumers, renewable companies, electric utilities, financial institutions, and investors.

A November 2015 Special Report in *The Economist*, having pointed out that French households pay about half as much as German ones for electricity, commented[2]:

Germany has made unusually big mistakes. Handing out enormous long-term subsidies to solar farms was unwise; abolishing nuclear power so quickly is crazy. It has also been unlucky. The price of globally traded hard coal has dropped in the past few years, partly because shale-gas-rich America is exporting so much. But Germany's biggest error is one commonly committed by countries that are trying to move away from fossil fuels and towards renewables. It is to ignore the fact that wind and solar power impose costs on the entire energy system, which go up more than proportionately as they add more.

In 2021 Germany's greenhouse gas emissions were 38% below 1990 levels, short of the 2020 abatement target of 40%.

The Russia-Ukraine war prompted a reappraisal of the plan to close the country's three remaining reactors at the end of 2022. The German coalition government, following a 'stress test' conducted by the country's power grid transmission operators, agreed in September 2022 to keep Isar 2 and Neckarwestheim 2 on standby until mid-April 2023. In October 2022 Chancellor Olaf Scholz announced that the country would keep the three remaining nuclear power reactors in operation until mid-April 2023.

Prior to the September 2022 decision, the Green party had been opposed to extending the operation of any reactors. The FDP argued that new fuel should be ordered so that the country's remaining reactors could operate into 2024.

Legal claims following March 2011

The country's four nuclear power utilities pressed claims for compensation and in particular were suing the government over continuing with the nuclear tax introduced in relation to the 8- and 14-year licence extensions agreed in September 2010. Claims for compensation were also on the basis of write-down of plants,

cancelled upgrades which were in train following the September 2010 policy change, and decommissioning costs brought forward. While RWE and E.ON are public companies, Vattenfall is owned by the Swedish government, and EnBW 46.55% by the Baden-Württemberg government, then a Social Democrat-Green coalition. Another 46.55% of EnBW is owned by the state's municipalities.

In March 2021 German power companies and the government announced that they were satisfied with an agreement reached. Under the agreement the power companies will receive a total of €2.4 billion: Swedish company Vattenfall will receive €1.4 billion, RWE €880 million, EnBW €80 million and E.ON €42.5 million. Vattenfall will also receive €188 million for the sale of nuclear plant production quotas to E.ON. In exchange for the settlement, the power companies agreed to drop all proceedings connected to Germany's nuclear phase-out.

Nuclear fuel tax

In September 2011 the government's continuing tax on nuclear fuel was rejected by the Hamburg Tax Court. The Court expressed "serious doubt" that the nuclear fuel tax was compatible with the German constitution. It granted a request from E.On to refund some €96 million, and nuclear fuel tax collections were to be suspended. The first lawsuit had been brought by EnBW, which had paid the tax when it refuelled a reactor in July and quickly launched legal action, claiming the tax was unconstitutional and contrary to EU law. The court's judgement said that the tax does not qualify under the constitution as a consumption tax, and anyway those should not be applied to single-purpose supplies like nuclear fuel. The court took its decision based on these constitutional points and did not consider other areas the utility had contested: whether the tax violated equality laws or EU directives on taxation. In October RWE and E.ON were refunded €74 and €96 million respectively. However the government then challenged the ruling and resumed collections of the tax.

In January 2013 the Hamburg Tax Court ruled more definitely that the German tax on nuclear fuel is simply "to siphon off the profits of the nuclear plant operators" and therefore unconstitutional. It referred the question to the Federal Constitutional Court and the EU Court of Justice (ECJ). E.ON, RWE and EnBW said the tax, of which they have paid about €5 billion, is illegal and favours other electricity sources, and have called for the tax to be repaid.

Since January 2011, E.ON had paid €2.3 billion in nuclear fuel taxes, EnBW had paid €1.1 billion, and RWE had paid €1.6 billion by the end of 2015, as well as bearing much greater costs with reduced revenue from the government's policy U-turn in March 2011. In April 2014 the Hamburg Tax Court upheld a demand from nuclear operators to refund about €2.2 billion, on the basis that the tax was a levy on profits and unconstitutional. But the court also allowed the matter to be referred to the Federal Fiscal Court (in addition to the cases pending at the Constitutional Court and the European Court of Justice).

In a non-binding preliminary opinion in February 2015, the EU Court of Justice found that the German nuclear fuel tax on utilities "that will be used to pay for decommissioning power reactors in the country" was legal, and that it did not violate EU taxation rules on electricity. Then in June 2015 judges at the Luxembourg-based Court of Justice of the European Union (ECJ) ruled "that EU law does not preclude a duty such as the German duty on nuclear fuel." The court also said the duty on nuclear fuel did not constitute illegal state aid to non-nuclear sources. In June 2017 the Federal Constitutional Court ruled that the nuclear fuel tax was "formally unconstitutional and void", which meant that the three utilities were to be reimbursed some €6.3 billion paid between 2011 and 2016 – €2.8 billion to E.ON, €1.7 billion to RWE and €1.44 billion to EnBW, plus interest.

An extended comment on the legal situation by a German energy law specialist was published by World Nuclear News (10 June 2015).

Earlier in March 2014 E.ON announced to BNetzA that its 1275 MWe Grafenrheinfeld nuclear power plant in Bavaria would close earlier than December 2015, due to the fuel tax of some €80 million making it uneconomic to refuel for that last period. In June 2015 when it closed it had operated 33 years.

Other lawsuits

Apart from contesting the fuel tax, all the nuclear generators sought compensation for the effective confiscation of generating rights from the eight reactors ordered shut after March 2011, despite safety assurances from the regulator as noted above.

RWE filed a lawsuit against the government regarding closure of its Biblis-B and said that the phase-out cost the company over €1 billion in 2011 alone. In February 2013 the administrative court in

Hesse found that the government had acted illegally in ordering the closure of Biblis A and B in March 2011. In January 2014 the German Supreme Administrative Court endorsed this by ruling that the forced closure of the Biblis plant by the state was "formally unlawful because [RWE] had not been consulted and this constituted a substantial procedural error." Biblis A and B, total 2407 MWe net, had been licensed to operate until 2019 and 2021 just two months before the shutdown order.

E.ON was also seeking €8 billion in compensation. In July 2016 a regional court in Hannover ruled that the company was not entitled to €382 compensation for the early closure of Isar 1 and Unterweser units. The decision was based on EnBW's failure in April in a Bonn court, and due to the company's failure to seek immediate legal action against the moratorium.

Vattenfall in June 2012 contested the confiscation of generation rights for the Brunsbüttel and Krümmel nuclear power plants, and filed the case with the autonomous International Centre for Settlement of Investment Disputes (ICSID) in Washington, which was designed in 1965 by the World Bank and established by a convention now signed by 143 countries. It had previously said simply that it expects full compensation for its costs, which it noted as SEK10 billion ($1.5 billion) for the first half of 2011 alone. In mid-2013 it announced a SEK 10.2 billion (€1.2 billion) write-off on those two plants. In October 2014 the energy minister said Vattenfall was seeking €4.7 billion compensation, the company saying that this was based on the Energy Charter Treaty which provides security to corporate investments against political risks. The ICSID opened its hearing in October 2016, before being suspended in March 2021.

In November 2020 a German court ruled in favour of Vattenfall in its case against the government relating to the Brunsbüttel, Krümmel and Mülheim-Kärlich nuclear plants. The Swedish utility argued that the conditions for compensation payments to producers impacted by the country's phase-out policy were not clearly regulated, and that previous amendments to the law could lead to a reduction of claims. The court ruled that the government must revise a 2018 draft law covering compensation for nuclear plants because it did not meet guidelines set out by the court in 2016, when it warned that their closure might violate certain property rights.

EnBW supported the legal actions brought by the other utilities, saying that the government's actions infringed its property rights. By May 2014 it had paid €790 million in the fuel tax for its two closed reactors. In December 2014 EnBW said it would file suit

against the federal and state governments, on the same basis as RWE, which was awarded €235 million (which has been appealed by the Hesse state government). It sought compensation of €261 million, but a regional court in Bonn ruled in April 2016 that the claim could not be allowed to stand because EnBW had not immediately used “all legal means available” to avert having its two reactors – Neckarwestheim 1 and Phillippsburg 1 – shut down.

In March 2021, it was announced that an agreement had been reached between the power companies and the German government (see above). The €2.4 billion to be paid is substantially below that sought by the companies.

Earlier in October 2015 *Reuters* reported:

Since Fukushima, shares in Germany's top three energy groups – E.ON, RWE and EnBW – have lost an average 56 percent, or €50 billion in combined market value, while racking up €65 billion in net debt, about twice their current combined market value. They have filed lawsuits against the government, claiming more than €24 billion related to Merkel's nuclear policy, which they claim is unfair and has rid them of one of their main profit centres overnight.

In February 2017 RWE said Germany's nuclear energy phase-out fund had imposed a "substantial one-off burden" on its business last year. The utility will pay €6.8 billion by July to indemnify itself from "largely politically induced disposal risks and avoid a high, disadvantageous interest burden." The charge includes a €1.8 billion so-called risk premium. Subject to parliamentary approval, the nuclear utilities will together pay a total of €23.6 billion into the fund, including a €6.2 billion risk premium. Payment must be made no later than 2026. They will then be cleared of any responsibility for final disposal of used fuel.

Transmission and supply implications

The German federal network agency and grid authority, Bundesnetzagentur (BNetzA), reported at the end of May 2011 on the implications of plans to close down nuclear generation. It strongly warned of resulting vulnerability to major failures and also unreliability especially in the south. Grid stability was the major concern, along with generation and transmission capacity.

In May 2012 Germany announced plans to upgrade and expand its electricity grid over the next decade in order to help renewable energy sources fill the gap left by its phase-out of nuclear power. At

the request of the government, the country's four grid operators (TSOs) – 50Hertz, Amprion, TenneT TSO and TransnetBW – drew up a joint network development plan which identified the necessary grid expansions. The plan – co-ordinated by the federal network agency, Bundesnetzagentur (BNetzA) – would provide a basis for delivering the country's energy transition. State governments agreed to let the BNetzA coordinate plans, rather than asserting regional interests (though Bavaria later reneged on this).

A bill introduced to the Bundestag in March 2013 identified 36 transmission projects costing some €10 billion as high priorities. The government wanted to reduce the timeframe for new power lines to four years on average, and the Federal Administrative Court would handle any legal cases arising from the power line developments, a measure to speed up the projects. Previously lawsuits could be brought in local or regional courts. Meanwhile Germany depends on neighbouring countries to route its power from north to south. The Czech government in 2012 complained it was close to a blackout because the German wind farms overloaded its grid. Early in 2014 the Bavarian government called for a moratorium on TenneT's and TransnetBW's SuedLink proposal linking Schleswig-Holstein in Germany's north to connect with the southern grid at the Grafenrheinfeld nuclear plant which closed down at the end of June 2015. This is near Schweinfurt in northern Bavaria.

More broadly, onshore high-voltage grids in Germany are undergoing considerable expansion to facilitate *Energiewende* and the development of the European electricity market. The network upgrades and additions required were expected to reach a cost of €20 billion by 2022. The four TSOs estimated that expanding wind power on the North and Baltic Seas would cost another €12 billion, TenneT expects to invest at least €22 billion by 2025 (not all in Germany), and another of the transmission companies estimates its own costs until 2025 to be €10 billion. While these investments "account for only a fraction of the cost of the energy transition, much success depends on their implementation." Later estimates in 2022 from the German Federal Network Agency (*Bundesnetzagentur*) anticipated added costs of about €55 billion by 2030 for the expansion of the high voltage network.

Failure to upgrade the electricity transmission grid would cause higher costs elsewhere. In May 2016 BNetzA put the cost of the required 7000 km of new transmission lines at €35 billion, with priority given to the three north-south links. In 2019 the TSOs outlined extra spending of €18 to €27 billion on top of previous estimates needed to connect offshore wind farms to the grid.

Early in 2016 grid projects were broadly covered either by the energy network expansion law Energieleitungsausbaugesetz (ENLAG) of 2009 or by the 2015 federal transmission system needs act, BBPlG (Bundesbedarfsplangesetz). ENLAG aimed to expedite 22 urgent transmission projects identified by DENA, and nearly all of these were to be completed by 2020. Completion of the Thüringer Strombrücke line (or Südwest-Kuppelleitung) from Lauchstädt to Redwitz, at the end of 2015 was a major landmark for TenneT. Another 43 projects are identified in the BBPlG, based on the 2014 version of the Network Development Plan (NEP) presented annually by TSOs to the BNetzA. BBPlG projects are subject to accelerated planning procedures carried out by the regulator, and BBPlG brings legal force to a mid-2015 decision to prioritize underground cabling of HVDC cables over overhead lines, where previously the opposite had been the case. The change arose largely from Bavarian opposition to overhead lines. In October 2015 the government approved plans for about 1000 km of high-voltage transmission lines from the north and close to populated areas to be built underground. The energy ministry estimated that the underground option would cost €3 to 8 billion more than overhead lines, to be added to consumers' bills, but was expected to speed up approvals. TenneT warned of cost and schedule delays to the SuedLink project (corridor C). Early in 2017 the EC approved €40 million for a study on "urgently needed" Suedlink on two routes: Brunsbuettel-Grossgartach and Wilster-Grafenrheinfeld.

In February 2017 the four TSOs reported to BNetzA on demand for redispatch measures to secure grid stability and concluded that 2.1 GWe of new fast-response open cycle gas turbine units in Bavaria, Baden-Württemberg and the southern region of Hesse was needed by 2021 to counter wind growth and nuclear decline, with 2020 and 2025 identified as exceptionally critical for grid stability.

Maintaining grid stability in 2021 cost €1.4 billion, due to redispatch* where prioritized renewable power causes transmission congestion and conventional power stations are paid to reduce output.

* Redispatching is an intervention in the market-based operating schedule of generating units in order to shift feed-ins from power stations. Based on the contractual obligations of TSOs, power stations are instructed to reduce their feed-in power, while other power stations are simultaneously instructed to increase their feed-in power. Redispatching is used by network operators to ensure the safe, reliable operation of electricity supply networks. It is carried out to prevent power lines becoming overloaded or to relieve

overloading on power lines. The network operator must reimburse the power stations participating in the redispatch for the costs they incur.

Replacing and closing down conventional generating capacity

Bundesnetzagentur (BNetzA) has received numerous requests from operators to retire coal- and gas-fired plants which have become unprofitable, and it has approved many of these as new coal-fired capacity comes online. However, particularly in the south, plant closures have exceeded new capacity coming online. E.ON's 1275 MWe Grafenrheinfeld nuclear reactor was closed in mid-2015. This gave rise to a net reduction of southern capacity of 1.7 GWe. This southern capacity deficit has been exacerbated by the subsequent closures of Philippsburg 2 (1.4 GWe) and Gundremmingen C (1.3 GWe).

In March 2015 E.ON and co-owners applied to BNetzA to close down two state-of-the-art almost new CCGT plants, Irsching 4&5 (550 & 846 MWe) in southern Germany from April 2016. They have thermal efficiency of about 60%, and had been running under an arrangement which was break-even, but due to the increase in subsidized renewables' output and low wholesale power prices, the two CCGTs had "no prospect of operating profitably when the current contract with the network operator expired in March 2016," the owners said. Nevertheless in September 2015 TenneT TSO prohibited the planned closures by declaring the units to be system-relevant (as Irsching 3 and another E.ON unit have been), so that they therefore needed to be kept operational though run at a loss. In the event the two units were mothballed in April 2016, before being restarted in October 2020 after being contracted by TenneT to help stabilize the grid. The *Bundesverband der Energie- und Wasserwirtschaft* (BDEW) has said that the economic viability of more than half of Germany's planned power plants has been called into question by government policies.

In 2019 the German government announced that it would phase out electricity production from coal by 2038. In June 2020 the country opened a 1100 MW coal power plant, Datteln 4. This plant, expected to be the last to open in the country, was in June 2020 one of 84 coal-fired plants generating electricity in Germany. Fortum, parent company of Uniper, the plant's operator, said: "The launch of the power plant has given rise to much debate, due to fears that the plant will increase emissions from electricity

production in Germany. We understand people's concerns and we agree that coal must be phased out and emissions must be reduced. However, the transition to a low-emission society must be made without compromising security of supply or an affordable cost of energy, in a socially just manner. This has been the starting point for the comprehensive solution of the German government, which allows the commissioning of Datteln 4 and systematic phasing out of coal by 2038."

Economic and CO2 implications of nuclear policy changes

Fuelling the earlier dispute within the grand SDP-Green coalition government then in power, a January 2007 report by Deutsche Bank warned that Germany would miss its carbon dioxide emission targets by a wide margin, face higher electricity prices, suffer more blackouts and dramatically increase its dependence on gas imports from Russia as a result of its nuclear phase-out policy, if it were followed through. The Economy Minister and utility owners called for urgent review of the policy. The Bank estimated that 42 GWe of new generating capacity would need to be constructed by 2022 if shutdowns proceeded.

In May 2007 the International Energy Agency warned that Germany's decision to phase out nuclear power would limit its potential to reduce carbon emissions "without a doubt." The agency urged the German government to reconsider the policy in the light of "adverse consequences." It warned that if Germany both continued with its nuclear phase-out policy and maintained carbon emissions reductions, by about 2020 it would need to depend on some 25,000 MWe of base-load electricity capacity across its borders. The country already has significant interconnection with France, Netherlands, Denmark, Poland, Czech Republic and Switzerland. Connection with Russia's Kaliningrad Baltic exclave, where a 2400 MWe Russian nuclear plant was planned, was envisaged and Russia expected to export half the output of that plant to Germany until confronted with political realities which caused the Baltic plant construction to be put on hold.

However, in September 2010 then March and May 2011 as described above, policy changed again twice, and in September 2011, a study from KfW Bankengruppe, which supports domestic developments, said that about €25 billion per year would be required to meet the government's *Energiewende* nuclear phase-out goals. It put the total capital investment at €239-262 billion by

2020. This included up to €10 billion on fossil fuel plant, €144 billion on renewables plant and up to €29 billion on 3600 km of high-voltage transmission lines. The bank noted that large capital-intensive projects have a tendency to go over budget.

In February 2013 the Federal Minister for the Environment, Nature Conservation and Nuclear Safety said that the costs of *Energiewende* by the end of the 2030s could reach €1000 billion. Feed-in tariffs subsidizing renewables alone would cost some €680 billion by 2020, and that figure could increase further if the market price of electricity fell, he warned.

The wholesale electricity price is based on marginal cost pricing, and with the output from wind and solar PV being often virtually zero marginal cost, increasing proportions of these has driven down average wholesale prices since 2008. Hence many power stations with higher marginal costs are displaced from the market by merit-order effect, and this has been seen most acutely with gas-fired plants, where capacity factors in 2018 ranged between 6% and 23%. Coal-fired plants require more EU ETS emissions certificates, but while these have been cheap it is more economic to keep these coal-burners in operation in defiance of *Energiewende*.

The retail picture is in contrast to wholesale electricity prices. The prices of electricity for private and most commercial customers have risen sharply as *Energiewende* took hold. Early in 2016 the price for private households was more than 90% above the average level of 2000, due largely to the EEG surcharge or *Umlage* which now comprises 21% of the total, adding to taxes comprising 23% of the total. Over 2005-14 residential electricity prices in Germany increased by more than the average total residential cost in the USA. In the first half of 2020 electricity prices for household consumers in Germany were the highest in Europe.

Germany's decision to shut its nuclear plants means that back-up for its massive investment in intermittent new renewables needs to be from coal and gas, which was estimated to create an extra 300 million tonnes of CO_2 to 2020. Those emissions virtually cancel out the 335 Mt saving across the entire European Union that were intended to be brought about by the 2011 Energy Efficiency Directive from the European Commission. But *Energiewende* locks Germany into long-term dependence on lignite and black coal for dispatchable capacity, contrary to a major aspect of the popular sentiment driving that policy, and its predecessors.

In 2017 the government faced the question of a carbon floor price made acute by impending elections and coalition disagreement on

the matter. The CDU/CSU major party was concerned about high energy costs and prioritized grid expansion, while the minor party SPD was keen to have a carbon price. The French government promotes a €30/t CO_2 carbon price. While this would have a small effect in France, a floor price of €30/t would increase German wholesale prices by €15/MWh to about €50, according to Poyry.

German generating costs 2021 (source: Fraunhofer Institute)

Source	€ cents/kWh	Full-load hours per year	Consequent capacity factor (%)
PV	3.5 - 5.8	950-1300	11-15
Onshore wind	4.0 - 8.2	1800-4500	20-38
Offshore wind	7.3 - 12.0	3200-4500	38-52
Biogas	8.5 - 17.1	5000-7000	57-80
Lignite	10.4 - 15.1	6450-7450	74-86
Black coal	11.0 - 20.0	5350-6350	62-73
CCGT	7.9 - 13.0	3000-4000	34-46
Gas	11.5 - 29.0	500-2000	6-23
Household retail price	32.2		

Electricity from renewable energy, feed-in tariffs

As Germany's attitude to nuclear energy became ambivalent, policies were adopted to promote renewable sources, notably solar and wind, though Germany is not well placed geographically in relation to either. Such policies are primarily to reduce carbon dioxide emissions. Due to the feed-in tariffs of the Renewable Energy Sources Act (EEG – *Erneuerbare Energien Gesetz*) passed in 2000, wind power has become the most important renewable source of electricity production in Germany.

Renewable electricity fed into the grid was paid for by the network operators at fixed feed-in tariffs (FITs), the costs being passed onto electricity consumers, so that there are no subsidies by the government itself. The tariffs are different for specific technologies and subject to a reduction of about 5% each year as an incentive for price reductions in new plant. The price is guaranteed for 20

years after completion of the plant, so that the operators have confidence in their planning criteria.

The coalition parties in the new government from late 2013 agreed to reduce the capacity targets from those set in 2010 and to revise the EEG law to reduce subsidies for renewable energy projects (see below). It appears that this will take priority over reforming the EU's Emissions Trading Scheme (ETS), which Germany took a lead role in establishing.

Since 2013 Germany has on numerous occasions had negative spot power prices due to reduced demand and windy weather. Reduced demand from 2020 during the coronavirus pandemic exhibited the inflexibility of the country's generation system, with wholesale prices repeatedly turning negative.

A Finadvice report in July 2014 said that the lessons learned from the German *Energiewende* included:

1. Policymakers underestimated the cost of renewable subsidies and the strain they would have on national economies.
2. Retail prices to many electricity consumers increased significantly, more than doubling 2000 to 2013.
3. The rapid growth of renewable energy reduced wholesale prices in Germany, with adverse consequences on markets and companies.
4. The wholesale pricing model changed as a result of the large renewable energy penetration, now reacting to the weather.
5. Fossil and nuclear plants are facing stresses to their operational systems as they are now operating under less stable conditions.
6. Large-scale deployment of renewable capacity does not translate into a substantial displacement of thermal capacity.
7. Large-scale investments in the grid are required.
8. Overgenerous and unsustainable subsidy programmes resulted in numerous redesigns of the renewable support schemes, which increased regulatory uncertainty and financial risk for all stakeholders in the renewable energy industry.

Revised Renewable Energy Sources Act 2014, and subsequent revisions

Following the September 2013 elections, the CDU-led government pledged to reform the 2000 Renewable Energy Sources Act (EEG – *Erneuerbare Energien Gesetz*), diminishing the reliance on feed-in tariffs for new solar and wind power output and favouring dispatchable generation which can respond to demand. The Federation of German Industries (BDI) and other industry groups

had been lobbying for a curb on feed-in tariffs, and household consumers were being hurt by high prices. This also raised the possibility of shifting some of the cost burden onto industries which had been exempt from the EEG surcharge or *Umlage*.

After consultations with 16 states, the federal government in April 2014 announced draft revisions of the EEG to limit energy price rises. The new law would hold the EEG surcharge at 6.24 c/kWh through to 2017, the renewable energy caps announced earlier were confirmed: offshore wind 6.5 GWe by 2020 and 15 GWe to 2030, onshore wind 2.5 GWe net added per year, solar PV also 2.5 GWe per year added. The caps are designed to allow about 11 TWh renewables growth each year. Renewables support continues to be granted for a 20-year operating period, albeit at much lower rates after the first five years.

Except for small plants, most renewables power sales are to be by 'direct marketing' by generators, with revenue supplemented by premiums calculated as the difference between the fixed feed-in tariff and the average wholesale price of electricity. The new arrangement is in place of feed-in tariffs, which the EC had ordered to be phased out over 2016-20. The new law took effect in August 2014.

One major issue is whether industry on-site power generation should be subject to the EEG surcharge. Some 50 TWh/yr is now generated by individual industry autoproducers to ensure reliability of supply, about 25% of the power used in industry. In the draft act, established autoproducers continued to be exempt, as were businesses which are fully independent of the grid, but other industry sources will pay 50% of the 6.24 c/kWh, or 15% in certain situations. This exemption was changed in the amended legislation after EC involvement. Other changes included reduced subsidies for renewables, and from 2017 those sources had to compete.

At the end of 2015 the EEG was again revised for 2016 onwards, with renewables limited to 45% in 2025 and 60% in 2035 in order to synchronize with network expansion, to secure planning and development of the conventional (fossil and nuclear) power station fleet and so that Germany's neighbours can adapt their own electricity systems to predictable renewable energy additions. The proposals for EEG 2016 say support for onshore wind, offshore wind and large PV plants with more than 1 MWe will be fixed in an auction system from 2017, to cover 80% of renewables generation produced in newly-installed plants each year. For other plants, no change from EEG 2014.

The process of revising the EEG was carried out as part of a broad consultation on the future electricity market, referred to by the Federal Ministry for Economic Affairs and Energy (BMWi) as 'Electricity Market 2.0'. This commenced with the publication in October 2014 of a green paper, followed in July 2015 by a white paper titled An electricity market for Germany's energy transition. This led to the new Act on the Further Development of the Electricity Market (*Gesetz zur Weiterentwicklung des Strommarktes*), which was passed by both houses of parliament in July 2016, heralded by the energy minister as "the biggest reform of the power market since the liberalization in the 1990s." Also referred to as the Electricity Market Act (*Strommarktgesetz*), it establishes the primacy of energy-only markets and replaced politically-set feed-in tariff support with competitive tenders for renewables. The legislation reinforces the central role of the wholesale power market by allowing uncapped scarcity pricing for electricity, and outlines various capacity reserves to assist security of supply while reducing sector emissions of CO_2. These reserves include the provisions of the June 2013 regulation on reserve power plants to counter the transmission restriction from north to south, and new capacity and security (lignite) reserves.

Whereas feed-in tariffs were set differentially between the north (more wind) and south (most demand), the new auction system does not allow that, so favours the north.

Electricity from new coal-fired plants

When EU ETS carbon prices are low, coal is more profitable than gas, and there is an incentive to use lignite, despite its higher CO_2 emissions.

An insight on the continued reliance on lignite can be gained from RWE, which in 2012 commissioned BoA units 2&3 in North Rhine-Westphalia near Cologne, 2200 MWe billed as "the world's most advanced lignite-fired power station" and costing €2.6 billion. Each unit can drop from full power by 500 MWe in 15 minutes and then recover as required, "demonstrating the power station's ability to offset the intermittency of wind and solar power." RWE said: "BoA 2&3 is an important element of our strategy, for modern coal and gas-fired power stations are indispensable. Unlike wind and solar sources, they are highly flexible and capable of producing electricity 24/7, which makes them the trump card of energy industry transformation." The state premier said that the plant was "an important contribution to security of supply."

In July 2015, after months of intense negotiations, the government scrapped its proposed levy on coal-fired plants and resolved that as part of the revised capacity mechanism regime within the Electricity Market 2.0 some 2.7 GWe of lignite-powered generating capacity (representing about 13% of installed lignite power) would be gradually transferred to a 'security standby reserve' (*Sicherungsbereitschaft*). This would be over 2016 to 2020 by negotiation with RWE (1.5 GWe), Vattenfall (1.0 GWe) and Mibrag. This capacity would be brought online when needed and then progressively shut down after four years. The three utilities would receive €230 million per year compensation over seven years, total €1.6 billion. The EC approved the arrangements in May 2016 under state aid rules.

In June 2020 the country opened a 1100 MW coal power plant, Datteln 4. This plant, expected to be the last to open in the country, was in June 2020 one of 84 coal-fired plants generating electricity in Germany.

In Germany, 178 million tonnes of lignite was mined in 2014. To achieve this, 879 Mt of overburden was removed, so total earthmoving on one year was 14 times that for building the Suez canal. In 2020, Germany produced 30% of the lignite mined worldwide. The heat value of German lignite ranges from 7.8 to 11.3 MJ/kg, and has around 50% water content. It is used almost entirely for electricity production domestically or in nearby countries, though some is used for industrial heat. RWE is the largest lignite power producer. Pulverized lignite (LEP) has water reduced to about 11% and correspondingly higher calorific value, so is increasingly traded for industrial heat applications and municipal CHP plants. Vattenfall is a leading player in this along with Mibrag Mining Corporation which rails lignite up to 400 km, and Rheinbraun Brennstoff which supplies a Swiss cement factory 600 km away with lignite.

Germany's last black coal mine closed in December 2018, though black coal imports continue and new lignite mines are being opened.

Events in 2022 forced the German government to outline plans to burn more coal for a “transitional period”. In July 2022 Chancellor Olaf Scholz announced that temporary plans to reactivate coal and oil-fired power plants were in place to relieve energy shortages resulting from the war in Ukraine.

Uranium mining

From 1946 to 1990, some 220,000 tonnes of uranium (260,000 t U_3O_8) was mined in the former GDR, in Saxony and East Thuringia, notably at Wismut, with substantial environmental damage. Much of this was used in Soviet weapons programs, and for fuel in Eastern Europe. In 1991, 1207 tU was produced, in 1992: 232 tU and thereafter small amounts resulting from decommissioning and mine closure activities.

A small mine. Ellweiler, operated in West Germany 1960-89. All uranium is now imported, from Canada, Australia, Russia and elsewhere.

Fuel cycle

Annual demand for enrichment in 2021 was expected to be about 0.4 million SWU, most of which was provided by Urenco's Gronau plant, with capacity of 3.9 million SWU/yr.

Most of the depleted uranium tails from the Gronau plant have been sent to Novouralsk in Russia for re-enrichment, but these arrangements finished in 2010. Over 2007 to 2009 Urenco sent 6500 t of tails assaying 0.30% U-235 to Novouralsk for re-enrichment, and 402 tonnes assaying 0.235% to Eurodif in France for re-enrichment. From Russia 270 tonnes of enriched uranium product was returned in this period.

In 2008 & 2009 Urenco shipped 518 tonnes of tails assaying 0.26% or less from Gronau to Areva's W Plant at Pierrelatte in France for deconversion. To the end of 2009, 1700 tonnes of UF_6 from Gronau had been deconverted there and returned to Gronau as U_3O_8.

Fuel fabrication is undertaken by Areva, mostly at Lingen in Germany.

Thirteen German reactors received licences to use mixed oxide (MOX) fuel (though some have now been shut down), using

plutonium recycled from spent fuel. A MOX plant at Hanau in Hesse was never allowed to operate, so all MOX fuel was imported.

Until 1994 utilities were obliged to reprocess spent fuel to recover the usable portion and recycle it. From 1994 to 1998 reprocessing and direct disposal were equally acceptable to the federal government, but the policy of the coalition government from 1998 to 2009 was for direct geological disposal of spent fuel and no reprocessing after mid-2005 (although firm contracts for reprocessing, totalling US$ 7.3 billion, were in place with BNFL and Areva).

Radioactive Waste – policies

In 1963 the federal government issued a recommendation to use geological salt formations for radioactive waste disposal. In 1973 planning for a national repository started, and in 1976 the Atomic Energy Act (AtG) was amended to make such disposal a responsibility of the federal government.

In July 2013 two acts were passed, the Repository Site Selection Act (StandAG) and another to establish a Federal Office for the Regulation of Nuclear Waste Management (*Bundesamt für kerntechnische Entsorgungssicherheit*, BfE) under the Ministry for Environment, Nature Conservation, Building and Nuclear Safety (BMU). The BfE regulates the site selection process and supports the ministry in relation to final disposal of radioactive waste. Its initial task was to ensure the refinancing of the site selection process, and then supervise site selection, evaluate proposals, certify that selection is properly undertaken according to the StandAG, oversee environmental assessment and present a final proposal for a site. Following the commissioning of the Konrad final repository for low- and intermediate-level waste and approval for decommissioning of Morsleben, it will grant relevant licences and permits to proceed with final waste disposal.

The federal government through the Federal Office for Radiation Protection (*Bundesamt für Strahlenschutz*, BfS) has been responsible for building and operating final repositories for high-level waste, but progress in this has been hindered by opposition from *Länder* governments. The BfS is responsible for licensing all nuclear waste transports.

In 2013 the federal environment ministry (BMU) announced that the federal government and all 24 states had finally reached agreement on drafting a repository law (see above), and that the

power utilities should spend €2 billion to find and develop a new repository. The industry body representing the companies responded that they were not prepared to do so, having already invested nearly that much in Gorleben. However, the new Repository Site Selection Act (StandAG) which was passed in July 2013 created a 33-member commission in May 2014 to develop 'basic principles' for site selection, including safety and economic requirements, and selection criteria for rock formations. The commission included representatives from parliament, academia, civil society organizations, industry, the environment and trade unions. Its initial recommendations offering a "comprehensive approach to responsible and safe disposal of all radioactive waste" were adopted by the cabinet in August 2015, with the plan to be submitted to the EC for approval. The commission's final report was submitted to the government in July 2016.

According to the commission's final report, the site with "the best safety" is to be determined in a three-phase process and defined by federal law. The site selection should be accompanied by extensive public participation with bodies at regional, inter-regional and national level. The repository could be located in salt, clay or crystalline rock. The commission said the "controversial" Gorleben rock salt formation in Lower Saxony has not been excluded in its report.

The report projects some 10,500 tonnes of used fuel from the operation of nuclear power plants, which could be stored in about 1100 containers. A further 300 containers of high- and intermediate-level waste are also expected from the reprocessing of used fuel, as well as 500 containers of used fuel from research and demonstration reactors. While the former Konrad iron ore mine in Salzgitter is favoured for low- and intermediate-level waste (see below), another as yet undetermined site for high-level waste remains to be identified. The German Atomic Forum (*Deutsches Atomforum*, DAtF), said: "In addition to the process and criteria, the commission has also developed a comprehensive and extremely ambitious involvement process that should give citizens, particularly in affected regions, far-reaching opportunities for participation. A consistent and targeted approach is needed to arrive at a solution to this long-disputed issue."

The Federal Company for Radioactive Waste Disposal (*Bundesgesellschaft für Endlagerung mbH*, BGE), was set up in July 2016 as a state-owned company under the BMU. It is headquartered in Peine. It is the designated project owner and operator of radioactive waste repositories following the Final Repository Commission's report. Since April 2017 the BGE has

been operating Asse II, Gorleben and Morsleben. It is responsible for site selection procedures for a final repository for heat-generating radioactive waste. In December 2017 BGE was merged with *Deutsche Gesellschaft zum Bau von Endlagern für Abfallstoffe mbH* (DBE), formerly a 75% subsidiary of spent fuel cask supplier GNS.

In December 2016 the Bundestag in a 581-58 vote resolved to create a €23.6 billion state-owned fund, the *Fonds zur Finanzierung der Kerntechnischen Entsorgung* (KENFO), to pay for the interim storage and disposal of all German used fuel and nuclear waste. The four nuclear utilities will provide the funding and will then have no further financial responsibility. The total includes a 35% risk premium in case costs are greater than anticipated. Earlier the energy minister said that the legislation "clarifies responsibility for nuclear waste. It ensures the long-term financing for decommissioning, dismantling and disposal without the transfer of costs to society or jeopardizing the economic situation of operators." It was reported that RWE and E.ON would pay €16.7 billion between them, Vattenfall €1.75 billion, and in March 2017 EnBW said it would pay €4.7 billion, including €2.4 billion risk premium. The fund had received €24.7 billion by August 2017 according to the Federal Ministry for Economic Affairs and Energy (BMWi), and is expected to grow to about €70 billion by 2100 through investment. The companies have already set aside some €38 billion for decommissioning their reactors – see section below.

Radioactive waste – responsibilities

The utilities have been responsible for interim storage of spent fuel, and formed joint companies to build and operate offsite surface facilities at Ahaus and Gorleben. Subsequent policy was for interim storage at reactor sites. In mid-2013 the licence for interim storage at Brunsbüttel was revoked, having been granted for 40 years in 2003. The facility was commissioned in 2006. In 2016 BfE gave EnBW permission to move 342 used fuel assemblies from the shutdown Obrigheim plant 50 km to interim storage at Neckarwestheim, to allow decommissioning at Obrigheim to proceed. In September 2016 Vattenfall was given permission by the state government to transfer 990 fuel assemblies from the storage pool at Krümmel into CASTOR dry storage casks onsite. It has applied to build a warehouse there for low- and intermediate-level waste which will later go to the Konrad repository.

GNS Gesellschaft für Nuklear-Service mbH (GNS) set up in 1977 and owned by the four nuclear utilities has been responsible for all

operations regarding the transport and disposal of waste in Germany, at nine sites. It also offers products and services outside Germany – it secured a €1.52 million contract with Sogin in Italy in 2015 for decommissioning and waste disposal. Its 75%-owned subsidiary *Deutsche Gesellschaft zum Bau und Betrieb von Endlagern für Abfallstoffe mbH* (DBE) constructed and operated repositories, notably Konrad and Gorleben, while decommissioning Morsleben. GNS developed and supplied the various types of CASTOR and CONSTOR casks for transporting and storing used fuel.

Following the December 2016 legislation, in March 2017 the BMU and GNS established the *Bundes Gesellschaft für Zwischenlagerung mbH* (BGZ) joint venture to enable the government to take over intermediate storage and final disposal of radioactive waste. In May 2017 GNS announced that it had reached agreement with the BMU for the transfer of its share in BGZ so that the federal government would become the sole owner of BGZ. As part of the agreement, GNS will transfer its interim storage activities to the government, including the existing central interim storage facilities in Ahaus and Gorleben which were transferred to BGZ at the end of July 2017. GNS said that 329 casks of HLW were at Ahaus and 113 casks at Gorleben (5 spent fuel, 108 vitrified HLW from Areva at La Hague). Some 80 GNS employees at both sites were transferred to BGZ, while around 70 GNS employees at its headquarters in Essen became responsible for the administration of BGZ. The management of 12 onsite interim storage facilities at German nuclear power plants was transferred to BGZ starting with HLW and used fuel in 2019, and 12 warehouses with ILW-LLW from operation and dismantling of nuclear power plants in 2020. “As a result, the responsibility for the interim storage of radioactive waste from energy supply companies will be centrally placed in the hands of BGZ” (BGZ website). The rationale for state takeover of all waste is that interim storage is likely to be for several decades, and the future of nuclear utilities and hence GNS is uncertain beyond the 2023 German phase-out.

In August 2021 France’s Orano contracted with the four German utilities to return all the intermediate-level waste it was holding, or its mass and activity equivalent to simplify the actual shipments. It results from reprocessing 5310 tonnes of German used fuel at La Hague over 1977-2008.

Radioactive waste – sites and operations

The last separated high-level waste from past reprocessing in France and UK is expected to be returned to Germany by the end of 2024 and stored. A total of 166 large casks of glass canisters will be involved, and following the last shipment from La Hague in November 2011, at least 50 of these are already in storage at Gorleben. Each holds 28 tonnes of vitrified HLW. A further 300+ casks with canisters of compacted waste from reprocessing could immediately go to a final repository, the canisters possibly into boreholes. In June 2015 the environment ministry announced a plan for some of this separated HLW, whereby 26 casks will be held at four interim storage sites. Five will be at Phillippsburg, and 21 at Biblis, Brokdorf and Isar nuclear power plants. These sites were selected as being "best placed from technical, legal and procedural aspects as well as from a political perspective." The German utilities – EnBW, E.ON, RWE and Vattenfall – welcomed the ministry's proposal and said they will now examine it in detail "with location, economic efficiency and inter-site aspects." E.ON said: "The four companies expressly declare their readiness to implement common solutions that can be legally approved, are economical and acceptable under corporate law and are legally secure."

A pilot reprocessing plant known as WAK (*Wiederaufarbeitungsanlage Karlsruhe Betriebsgesellschaft*) operated at Karlsruhe from 1971 to 1991, processing 206 tonnes of used fuel using the PUREX process. The separated HLW from this was 60 m^3 in liquid form, and after a series of political delays it was vitrified in 2009-10. The 122 canisters of vitrified waste are stored at Greifswald while awaiting disposal in a geological repository. The low- and intermediate-level waste from WAK were disposed of in the salt mine repository at Asse in Lower Saxony, and comprised about half of the waste emplaced there.

Gorleben: Following an exhaustive site selection process the state government of Lower Saxony in 1977 declared the salt dome at Gorleben to be the location for a national centre for disposal of radioactive waste. It is now considered a possible site for geological disposal of high-level waste. This would be for about 5% of total waste with 99% of the radioactivity. BGZ operates the Transport Container Storage (TBL) there, built in 1982-83 and now with 113 casks (5 spent fuel, 108 vitrified HLW from Areva at La Hague). It is licensed by BfE. The Gorleben waste storage facility is used to store radioactive waste with negligible heat generation. The licensing and supervisory authority for this and the Pilot Conditioning Plant there is the Lower Saxony Ministry for the Environment, Energy and Climate Protection.

The site could be available as a final repository from 2025, with a decision to be made about 2019. Some €1.5 billion was spent over 1979 to 2000 researching the site, and the investment in it from the power utilities now stands at about €1.6 billion. Work stopped in 2002 due to political edict, but in October 2010 the BfS on behalf of the federal government applied to resume studies and extend the operating licence to 2020. Lower Saxony allowed this, and in 2013 it agreed that Gorleben should not be ruled out in further considerations proposed then.

Other proposals are for a HLW repository in opalinus clay, which occurs in a number of places in Germany. In July 2009 new repository criteria came into force, replacing rules dating from 1983. Authorities may now license an HLW repository only on the basis of scientific demonstration that the waste will be stable in the repository for a million years. In addition, all HLW disposed of in any German repository must be retrievable during the entire period the repository is operated.

The **Ahaus** facility is used for storing intermediate-level waste, including some used HEU fuel from research reactors. In 2010 the BfS approved shipment of 951 used fuel elements from the Rossendorf reactor in 18 sealed containers to Mayak in Russia for reprocessing, on the basis of the Russian Research Reactor Fuel Return Program. Rossendorf, in east Germany, was closed in 1991.

The **Konrad** site (a former iron ore mine) was under development as a repository since 1975, and was licensed in 2002 for intermediate- and low-level waste disposal to 2022, but legal challenges were mounted. These were dismissed in March 2006 and again in April 2007. A construction licence was issued in January 2008. Konrad will initially take some 300,000 cubic metres of waste – 95% of the country's waste volume, with 1% of the radioactivity. DBE plans for it eventually to accommodate 650,000 cubic metres of waste from the operation and decommissioning of nuclear power plants as well as from industry, medicine and research. It was initally expected to be operational by 2014 with storage chambers on six levels from 800 to 1300 metres depth, but has been delayed. The August 2015 programme did not seek an extension to the Konrad repository licence, as previously proposed, due to local opposition. Hence another repository will be needed for the balance of intermediate- and low-level waste produced by 2023, when Germany's last nuclear power reactors were shut down under the government's nuclear phase-out policy.

About 200,000 cubic metres of mostly low-level waste “with negligible heat generation” is likely to be moved to Konrad, along

with about 100,000 cubic metres of waste from Urenco's Gronau enrichment plant.

At the start of August 2022 the Federal Company for Radioactive Waste Disposal (*Bundesgesellschaft für Endlagerung*, BGE) awarded construction firm Ed Züblin AG a contract to construct a radioactive waste reloading hall at Konrad. Construction is scheduled to begin during the first quarter of 2023, with completion planned for 2025. Low- and intermediate-level waste is scheduled to arrive and be repackaged in the facility from 2027.

The **Asse** salt mine repository, licensed by federal and state agencies in the 1960s and 1970s, is now closed. It received 47,000 cubic metres of low- and intermediate-level waste from 1967 to 1978. It is in poor condition and is seen to represent a failure of proper licensing process. The BfS decided in 2010 that the waste should be moved from it, and rejected an alternative of filling it with concrete to provide a stable matrix for the 126,000 drums there.

The salt dome repository at **Morsleben** in east Germany for low- and intermediate-level waste was licensed in 1981, re-licensed post reunification, and was closed in 1998. It has 36,754 cubic metres of low- and intermediate-level waste but is in poor condition and is being stabilised with concrete at a cost reported to be €2.2 billion. The waste will remain there.

Konrad, Asse and Morsleben are all in central Germany between Hanover and Magdeburg, Gorleben is about 100 km southeast of Hamburg. Ahaus is in western Germany.

Decommissioning

Five VVER-440 units at Greifswald were closed in 1990 following reunification (unit 6 was complete but did not operate). 235 unused fuel assemblies were sold to Paks in 1996. Unit 5 had a partial core melt in November 1989, due to malfunctioning valves (root cause: shoddy manufacture) and was never restarted.

Gundremmingen A BWR was shut down following an accident in 1977. High tension lines from the plant short circuited requiring rapid shutdown of the plant, which resulted in pressure relief valves flooding it with slightly radioactive water. Repairs and modernization were deemed uneconomic.

Two units of a four-unit VVER-1000/V-320 power station were under construction at Stendal, but halted in 1990. Unit 1 was about 85% complete.

Power and experimental reactors shut down to 2006

Reactor	Type	MWe net each	Years operating each	Shut down	Status
Greifswald 1-4	VVER-440/V-230	408	Up to 16	1990	Dismantled
Greifswald 5	VVER-440/V-213	408	0.5	11/1989	Dismantled
Gundremmingen A	BWR	237	10	1/1977	Dismantled
Grosswelzheim	Prototype BWR	25	1	1971	Dismantled
Jülich AVR	Experimental HTR	13	21	1989	
Kahl	Experimental BWR	15	24	1985	Site unrestricted
Kalkar KNK 2	Prototype FNR	17	13	1991	
Karlsruhe MZFR	Experimental PHWR	52	18	1984	
Lingen	Prototype BWR	183	10	1979	Safestor
Mülheim-Kärlich	PWR	1219	2	1988	Dismantling since 2004
Niederaichbach	Experimental GCHWR	100	1	1974	Site unrestricted
Obrigheim	PWR	340	36	2005	Dismantling since 2013
Rheinsberg	VVER-70/V-210	62	24	1990	Dismantled
Stade	PWR	640	31	2003	
THTR	Prototype HTR	296	3	1988	Safestor
Würgassen	BWR	640	22	1994	Dismantled 2014
Total: 19		4655			

Power reactors shut down from March 2011

Plant	Operator	Type	MWe net	Years operating	Shutdown	Status
Brunsbüttel (KKB)	Vattenfall	BWR	771	30	2007	Licensed decomm
Krümmel (KKK)	Vattenfall	BWR	1260	25	2009	Shutdown
Biblis A (KWB A)	RWE	PWR	1167	36	2011	Licensed decomm
Biblis B (KWB B)	RWE	PWR	1240	34	2011	Licensed decomm
Isar 1 (KKI)	E.ON	BWR	878	32	2011	Licensed decomm
Neckarwestheim 1 (GKN)	EnBW	PWR	785	34	2011	Licensed decomm
Phillippsburg 1 (KKP)	EnBW	BWR	890	31	2011	Licensed decomm
Unterweser (KKU)	E.ON	PWR	1345	32	2011	Licensed decomm
Grafenrheinfeld (KKG)	E.ON	PWR	1275	33	2015	Licensed decomm
Gundremmingen B (KRB-B)	RWE	BWR	1284	33	2017	Licensed decomm
Phillippsburg 2 (KKP)	EnBW	PWR	1392	35	2019	Licensed decomm
Brokdorf	Vattenfall	PWR	1410	35	2021	Shutdown
Grohnde	E.ON	PWR	1360	37	2021	Shutdown
Gundremmingen C	RWE	BWR	1288	37	2021	Shutdown
Neckarwestheim 2	EnBW	PWR	1310	34	2023	Shutdown
Isar 2	E.ON	PWR	1410	35	2023	Shutdown
Emsland	RWE	PWR	1335	35	2023	Shutdown
Total: 17			20,400			

E.ON equity: Isar 1 100%, Unterweser 100%, Krümmel 50%, Brunsbüttel 33.3%, Grafenrheinfeld 100%, Gundremmingen 25%.
RWE equity: Biblis 100%, Gundemmingen 75%.
Vattenfall equity: Brunsbüttel 66.7%, Krümmel 50%.
EnBW equity: Neckarwestheim 100%, Phillippsburg 100%.

In 2012 eight reactors were prematurely shut down by government edict, for political reasons. This meant that the contributions to their respective decommissioning funds were truncated, rather than being allowed to accumulate for a full 40 or more years. The four operators in 2015 had a total of about €38 billion reserves set aside for decommissioning and waste disposal.*

* E.ON €14.6 billion, RWE €10.25 billion, EnBW €7.66 billion, Vattenfall €1.6 billion, Krümmel €1.8 billion, total €36 billion at the end of 2013.

EnBW announced that its two reactors – Neckarwestheim 1 and Phillippsburg 1 – would be directly dismantled without any safestor period, and in May 2013 EnBW submitted applications to decommission and dismantle them. In 2016 it applied similarly for Phillippsburg 2. In February 2017 EnBW received a decommissioning and dismantling licence for Neckarwestheim 1; and in April 2017 the same for Phillippsburg 1 from the Baden-Württemberg environment ministry. Work began soon after and will take 10-15 years. The licence for Phillippsburg 2 was received just before its shutdown in 2019.

In late 2012 Vattenfall Europe submitted an application to decommission and dismantle Brunsbüttel, which had been closed since 2007, and in August 2015 it applied similarly for Krümmel, which had not run since 2009. Vattenfall started removing fuel from Krümmel in September 2016 to dry casks on site, and planned to start defuelling Brunsbüttel the following month. Dismantling would be undertaken over a 15-20 year period. It has written off SEK 10.2 billion (€1.2 billion) on Brunsbüttel and Krümmel.

In January 2017 E.ON's PreussenElektra received a decommissioning and dismantling licence for Isar 1, the first such licence since 2011. Work will begin in 2017 and is expected to take 15 years at a cost of about €1 billion.

PreussenElektra's Brokdorf plant was shut down at the end of 2021 and dismantling began in December 2024. A process to remove all 764 used fuel assemblies from Brokdorf's storage pools to CASTOR casks took place between February 2023 and February 2026.

RWE applied in August 2012 to decommission and dismantle the two Biblis reactors. The Hesse ministry of environment approved plans in March 2017, including for defuelling unit B from 2017. The work is expected to take about 15 years. In December 2016 RWE applied for a permit to decommission and dismantle Emsland when it closes in 2022.

Decommissioning of the 17 nuclear units operating to 2011 and six other commercial units (total 23) was expected to cost €48 billion. The federal government ordered a review of the four utilities' decommissioning provisions, and after it reported in October 2015 the government said that the companies concerned had made sufficient provisions to cover all of the costs and had done so in compliance with the relevant rules. Their combined assets of about €83 billion would cover the costs of decommissioning the power plants and disposing of radioactive waste, and the expert opinion did not point to any need for additional action to be taken beyond these steps. The expert opinion found that the €38.3 billion of provisions made by the companies was based on higher cost estimates than the international average. Nevertheless, in April 2016 the 19-member Commission on the Review of the Financing of the Nuclear Phaseout (KFK) called for utilities to provide an extra €23.3 billion "risk premium" and pay all provisions into a state-run fund

In May 2015 E.On and Vattenfall Europe (VENE) signed an agreement to cooperate on decommissioning "in order to make the decommissioning and dismantling process of their joint venture nuclear power plants as economical as possible." They said that the main objective of the agreement "is to incorporate experience, especially from the largely-completed dismantling of the E.On nuclear power plant in Stade, in the planning and implementation of the decommissioning of the VENE power plants."

Energiewerke Nord GmbH (EWN) is wholly-owned by the German government and is responsible for the decommissioning of publicly-owned nuclear facilities and for managing the resulting radioactive waste. In addition to decommissioning the Greifswald nuclear power plant and the Rheinsberg experimental reactor in eastern Germany following the country's reunification, EWN is also involved in decommissioning the AVR reactor, which is adjacent to the research centre at Jülich.

Research

From 1956 a number of nuclear research centres were set up in West Germany, and most of these as well as university institutes were equipped with research rectors. Most of these reactors are now shut down and the centres have changed their roles. In 2015 the nuclear expertise of *Forschungszentrum Jülich* (Jülich Research Centre) was merged with the Experimental Reactor Consortium (AVR) under state-owned Energiewerke Nord GmbH (EWN), with

the federal ministry of finance as a shareholder. Its focus is on nuclear power and associated activities. The new organization, with some 300 employees, will have the full range of expertise in nuclear decommissioning, dismantling and waste disposal gained at Jülich over the past five decades. Jülich is in North Rhine-Westphalia.

In 1960 a 16 MWe experimental nuclear power plant ordered in 1958 was started up. Then in 1961 the AVR (*Arbeitsgemeinschaft Versuchsreaktor*) 13 MWe experimental high temperature reactor at Jülich was ordered, with fuel as a pebble bed. It operated for over 750 weeks from 1967 to 1988, most of the time with thorium-based fuel.

The 300 MWe THTR (*Thorium Hochtemperatur Reaktor*) at Uentrop was developed from the AVR and operated 1985-88 also using thorium-based fuel. Fuel fabrication was on an industrial scale. Several design features made the AVR unsuccessful, though the basic pebble bed concept was again proven. It drove a steam turbine.

The 200 MWt (72 MWe) **HTR-modul** was then designed by Siemens/Interatom and licensed in 1989, but was not constructed. It had low-enriched uranium pebble fuel which was tested in the AVR. This design was part of the technology bought by Eskom in 1996 and is a direct antecedent of the pebble bed modular reactor (PBMR) and the Chinese HTR-PM.

During 1970s and 1980s Nukem manufactured more than 250,000 fuel elements for the AVR and more than one million for the THTR. In 2007, Nukem reported that it had recovered the expertise for this and was making it available as industry support.

A fast breeder reactor, the 17 MWe Kompakt KNK 2 was built by Siemens and ran from 1978 to 1991. The much larger SNR-300 was also constructed by Siemens in the 1970s but for political reasons was never commissioned. The 1500 MWe SNR-2 was designed by KWU but not built.

In East Germany a research institute opened in 1956 and its research reactor started operation the following year. The first East German power reactor, the 70 MWe Rheinsberg PWR (VVER 220/V210), was connected to the grid in 1966, operating until it was closed by political decision in 1990.

In 1969 Siemens and AEG merged their nuclear activities to form Kraftwerk Union (KWU). KWU developed a series of PWR units

culminating in the standardized 1300 MWe Konvoi design, of which only three were built (though six preceding ones were similar).

Through the 1990s Siemens-KWU with utilities worked with EdF and Framatome to develop the 1600 MWe EPR, marketed by Framatome ANP (formed from Framatome-Siemens nuclear merger), then Areva NP.

At Jülich, Urenco maintains a centrifuge development and manufacturing centre.

The European Commission's Joint Research Centre focused on nuclear energy is at Karlsruhe, in Baden-Württemberg near the French border. This is being upgraded with a new laboratory to "enable the JRC to continue carrying out state-of-the-art nuclear research... The new laboratory will also be instrumental for maintaining EU expertise and skills in the nuclear field by providing training and open access to students and researchers." It is funded by the Euratom Research and Training Program, with an emphasis on nuclear safety and security.

Regulation and safety

In 1955 the West German government established an Atomic Ministry (BfA) with strong European links. The Atomic Energy Act was promulgated in 1959 and is the core legislation relevant to licensing and safety. The Radiation Protection Ordinance, Nuclear Licensing Procedure Ordinance and six other ordinances support this.

The Federal Ministry of Environment (BMU) is the main national body involved with licensing and supervising nuclear facilities, and is supported by the Federal Office for Radiation Protection – Bundesamt fur Strahlenschutz (BfS). However, licensing of nuclear power plants and other facilities is actually done by the states, which are responsible for implementing federal laws. The BMU supervises this and can issue binding directives.

Under BMU, the Reaktorsicherheitkommission or Reactor Safety Commission (RSK) conducts safety reviews of nuclear power reactors.

Also under BMU, the Entsorgungskommission (ESK) or Waste Management Commission operates. However, following passage of the new waste repository law in mid-2013, a new independent regulator – the Federal Office for Nuclear Waste Disposal – wa established.

The BfS is responsible for construction and operation of nuclear waste facilities. Individual utilities are responsible for setting aside funds for waste disposal and decommissioning.

The Verband der Grosskessel-Besitzer e.V. was founded in 1920 as the federation of the owners of large boilers. VGB PowerTech e.V. (VGB) is the European technical association for power and heat generation and works in close co-operation with Eurelectric on the European level and with the corresponding energy and water industries association (BDEW) on the national level. It undertakes research relevant to nuclear plant safety.

Public Opinion

Following protests concerning nuclear power plants in the 1970s, notably against construction of a plant at Whyl, by the end of the decade German public opinion was turning against nuclear power and embracing the notion of energy from nature.

The background to this in Germany is the long-standing influence of romanticism with love of forests and religious or mystical regard for nature which carried through into the 20th century as a complex reaction to industrial capitalism. In the 1960s it became coupled with far-left activism which transferred across to the formation of the Greens, the world's first major environmentalist political party. The politics of anti-nuclear protest gained an appeal to middle-class Germans, by conflating anti-NATO missile sentiment from being in the front line of a feared World War III and transferring this to the excellent plants that produced a third of their electricity very cheaply, while promoting idealistic visions of wind and solar potential.

In 1986 the Chernobyl accident caused great concern in Germany and made the negative image worse, thus consolidating opposition to nuclear power. Green politics gained new momentum: 'Red-Green' coalitions of Social Democrats and Greens were formed in the German states and eventually, in 1998, gained representation at federal level. Anti-nuclear activism came to define the heart and soul of the environmental movement, expressing a foundational myth. Climate change then became the headline public issue for Greens, which complicated but did not counter negative perceptions of nuclear power’s clean energy credentials in the public mind.

So for nearly four decades German public sentiment has been split in relation to support of nuclear energy. A poll late in 1997 showed

that some 81% of Germans wanted existing nuclear plants to continue operating, the highest level for many years and well up from the 1991 figure of 64%. The vast majority of Germans expected nuclear energy to be widely used in the foreseeable future. The poll also showed a sharp drop in sympathy for militant protests against transport of radioactive waste.

After the crucial October 1998 election a poll confirmed German public support for nuclear energy. Overall 77% supported the continued use of nuclear energy, while only 13% favoured the immediate closure of nuclear power plants.

In November 1998 Germany's electric utilities issued a joint statement pointing out that achievement of greenhouse goals would not be possible without nuclear energy. A few days later the Federation of German Industries declared that the "politically undisturbed operation" of existing nuclear plants was a prerequisite for its cooperation in reaching greenhouse gas emission targets. Nuclear energy then avoided the emission of about 170 million tonnes per year of carbon dioxide, compared with 260 Mt/yr being emitted by other German power plants.

A poll early in 2007 found that 61% of Germans opposed the government's plans to phase out nuclear power by 2020, while 34% favoured a phase out. Another poll in mid 2008 (N=500) showed that 46% of Germans wanted the country to continue using nuclear energy; another 46% said they supported the nuclear phase out policy, and 8% were undecided.

Following the Fukushima accident, in September 2011 a GlobeScan survey showed 52% of Germans thought that nuclear power was dangerous and plants should be closed as soon as possible (compared with 26% in 2005), *i.e.* they supported the government phase-out policy, 38% supported continuing use of existing plants but no new build (47% in 2005), and 7% supported use with building more (22% in 2005). Hence 90% opposed building new nuclear plants (73% in 2005). In response to the proposition that Germany could almost entirely replace coal and nuclear energy within 20 years by becoming highly energy efficient and depending on power from sun and wind, 62% agreed and 26% disagreed.

In June 2012 a poll by the Institut für Demoskopie Allensbach asked: “Do you think the federal government took the right decision for Germany to phase out nuclear by 2022?” Here 73% agreed that it took the right decision, and 16% answered no.

An opinion poll commissioned by the German Atomic Forum (DAtF) and carried out by Forsa in September 2013 asked whether nuclear

power plants should be shut down as planned (or even earlier) "or should the effects on a secure supply of electricity and on costs for consumers and industry be considered prior to future shut downs?" Here 59% opted for a conditional approach and 39% for the unconditional approach. However, this is not considered to represent a change in the underlying antipathy.

An opinion poll commissioned by the DAtF and carried out by Forsa in April 2014 showed that 72% supported a unified European energy policy and 56% opposed Germany reviewing its energy policy goals, *i.e.* the nuclear phase-out, the limitation of lignite mining and the ban on shale gas extraction in the light of energy security of supply concerns raised by the Ukrainian political crisis.

A poll held by ARD-DeutschlandTrend in August 2022 found that 41% of those surveyed believed that the country should continue to use nuclear energy in the long term. Only 15% of those surveyed were in favour of the remaining nuclear reactors being shut down at the end of 2022 – as was planned under Germany's nuclear phase-out policy – with 41% supporting an extension of the operation of the units by a few months. Similarly, a poll conducted the same month by the online survey institute Civey on behalf of *Der Spiegel* found that 78% supported the continued operation of the last three German reactors until the summer of 2023, with 67% of those questioned in favour of operating the reactors for another five years.

Non-proliferation

Germany is a party to the Nuclear Non-Proliferation Treaty (NPT) as a non-nuclear weapons state. Its safeguards agreement under the NPT came into force in 1977 and it is also under the Euratom safeguards arrangement. In 1998 it signed the Additional Protocol in relation to its safeguards agreements with both IAEA and Euratom. It is also a member of the Nuclear Suppliers Group.

Notes & references

Notes

a. German installed capacity figures provided by the Federal Network Agency for Electricity, Gas, Telecommunications, Post and

Railway (*Bundesnetzagentur*) – a federal agency of the Federal Ministry of Economics and Energy (*Bundesministerium für Wirtschaft und Klimaschutz*, BMWK) – differ from those provided by the International Energy Agency. Its SMARD (*Strommarktdaten für Deutschland*, Electricity Market Data for Germany) website gives the following installed capacity figures at the end of 2020:

Total 223 GWe, comprising 60.7 GWe wind, 48.2 GWe solar, 43.5 GWe coal, 31.7 GWe natural gas, 14.7 GWe hydro (of which 9.4 GWe pumped storage), 8.1 GWe nuclear, 8.0 GWe biomass, 8.0 GWe other. [Back]

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